Wood Science

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Wood Science
(A08842)

Short Title:	Wood Science		
Faculty	Science and Computing		
Department	Science		
Cluster	Forestry		
Author	TKENT		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module is designed to provide the student with a basic understanding of wood as a raw material. The module will introduce the student to the nature of tree growth. The structure and properties of wood will be described in lectures and assessed in practicals. The student will gain an understanding of the role of wood characteristics in determining the use of wood. A key element of the module deals with the effect of forest management practices on the characteristics of wood, and how forest managers may modify wood quality through forest operations.

Programmes

FORS-0003	BSc in Forestry (WD_SFORS_D)	1 / 2 / M
FORS-0003	BSc in Forestry (WD_SFORS_D)	2 / 4 / E

Indicative Content

- Fundamental properties and characteristics of wood
- Tree growth and wood formation
- Juvenile wood and reaction wood
- Cell structures and composition in softwoods and hardwoods
- Wood chemistry and cell composition
- Identification of timbers
- Natural durability and timber degrade
- Moisture content, shrinkage and movement in wood
- Wood density and strength properties
- Energy properties of wood
- Effect of forest management practices on wood quality

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe aspects of tree growth, macro- and microscopic wood structure and chemical composition of wood.
2. Describe the variation in wood properties; density, strength, natural durability and the causes of variation.
3. Explain the effect of forest management operations on wood properties.
4. Complete basic laboratory procedures, alone and in a group, on softwood and hardwood timber samples and slides.
5. Use standard procedures to assess and compare wood properties, moisture content, anisotropy, basic and bulk density and strength.
6. Record and present laboratory experiment results, carry out calculations and draw conclusions.

Learning and Teaching Methods

- Lectures: will be used to describe the nature of tree growth and the characteristics and properties of wood.
- Laboratory Practicals: will be used to develop analytical skills and practical measurement techniques in assessing wood properties and identifying wood and wood cell types.
- Independent Learning: reading recommended reference books, supplementary papers and handouts, and viewing recommended web based material to consolidate knowledge of the module content.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3
Continuous Assessment	50%	
Practical	50%	4,5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	24	
Independent Learning	87	

Essential Material

- "WIT Moodle Wood Science Module." https://moodle.wit.ie/course/view.php?id=49959
- Haygreen, J.G., R. Shmulsky and J.L. Bowyer. _Forest products and wood science: an introduction_. 5th Edition. USA: Blackwell Publishing, 2007.

Supplementary Material

- Bowyer, J.L. and R.L. Smith. _The Nature of Wood and Wood Products_. CD-ROM. 1999. USA: University of Minnesota Forest Products Management Development Institute.
- Hoadley, B. _Identifying Wood: Guide to getting accurate results with simple tools_. USA: Taunton Press, 1990.
- Hoadley, B. _Understanding Wood: A craftsman's guide to wood technology_. 2nd Edition. USA: Taunton Press, 2000.

Requested Resources

- Room Type: Biology Lab
- Lecture Room: Loose Seated